

QUESTION 1

Scenario: A newly established educational institution plans to launch its official website. Before developing the website, the management wants to understand the fundamental concepts of the World Wide Web.

Task: Create a simple HTML webpage explaining the following concepts with suitable illustrations or diagrams:

- Internet Overview
- Fundamental Computer Network Concepts
- URL and its Structure
- Domain Name
- Web Browsers and Web Servers
- Working Principle of a Website
- Difference between Client-side and Server-side Scripting

ANSWER:

The Internet is a worldwide network that connects computers and other devices and allows them to communicate and share information. The World Wide Web is a service that operates over the Internet and allows users to access web pages through web browsers.

The webpage created for this question explains the basic concepts of the Internet, computer networks, URLs, domain names, web browsers, web servers, website working principles, and client-side and server-side scripting.

SAMPLE CODE:

HTML CODE:

```
<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8">

  <title>Web Fundamentals</title>

  <style>

    /* Basic styling for the webpage */

    body {

      font-family: Arial, sans-serif;

      max-width: 900px;

      margin: auto;
```

```
        padding: 20px;
/* Styling for headings */
h1, h2 {
    color: #1a2b4c;
}
/* Styling for information boxes */
.box {
    background: #f0f4f8;
    padding: 15px;
    margin: 10px 0;
}
/* Styling for tables */
table {
    width: 100%;
    border-collapse: collapse;
}
th, td {
    border: 1px solid #ccc;
    padding: 8px;
}
th {
    background: #1a2b4c;
    color: white;
}
</style>
</head>
<body>
    <!-- Main heading -->
    <h1>Web Fundamentals</h1>
```

<!-- Internet Overview section -->

<h2>1. Internet Overview</h2>

<div class="box">

The Internet is a global network of interconnected
computers that communicate using TCP/IP protocols.

</div>

<!-- Network Concepts section -->

<h2>2. Fundamental Computer Network Concepts</h2>

<div class="box">

Node: Device connected to a network

IP Address: Unique address of a device

Protocol: Rules for data communication

Bandwidth: Data transfer capacity

</div>

<!-- URL Structure section -->

<h2>3. URL and its Structure</h2>

<p>

<https://www.example.com:443/path/page.html?query=1>

</p>

<!-- URL components are represented using a table -->

<table>

<tr>

<th>Part</th>

<th>Example</th>

<th>Meaning</th>

</tr>

<tr>

<td>Protocol</td>
<td>https://</td>
<td>Method of data transfer</td>

<td>Domain</td>
<td>example.com</td>
<td>Website address</td>

<td>Path</td>
<td>/page.html</td>
<td>Resource location</td>

</table>

<!-- Browser and Server section -->

<h2>4. Web Browsers and Web Servers</h2>

<div class="box">

A web browser requests and displays web pages.

A web server receives requests and sends the required web resources to the browser.

</div>

<!-- Client-server communication diagram -->

<h2>5. Working Principle of a Website</h2>

<div class="box">

Browser (Client) → HTTP Request → Web Server

Browser (Client) ← HTTP Response ← Web Server

</div>

```

<!-- Client-side and Server-side comparison -->
<h2>6. Client-side vs Server-side Scripting</h2>
<table>
  <tr>
    <th>Client-side</th>
    <th>Server-side</th>
  </tr>
  <tr>
    <td>Runs in the browser</td>
    <td>Runs on the server</td>
  </tr>
  <tr>
    <td>Example: JavaScript</td>
    <td>Examples: PHP, Python, Node.js</td>
  </tr>
</table>
</body>
</html>

```

CODE EXPLANATION:

The code uses **HTML5** to create the structure of the webpage and **CSS3** to provide styling. Different headings, paragraphs, lists, tables, and `<div>` elements are used to present the web fundamentals clearly. CSS classes are used to style the information boxes, URL diagram, tables, and client-server flow. HTML comments are included to identify the major sections of the webpage.

OUTPUT SCREENSHOTS:

Screenshot 1: Internet Overview and Fundamental Computer Network Concepts

Web Fundamentals

1. Internet Overview

The Internet is a global network of interconnected computers that communicate using standardized protocols (TCP/IP) to share data, resources, and services across the world.

2. Fundamental Computer Network Concepts

- **Node:** Any device connected to the network
- **IP Address:** Unique numerical identifier for each device
- **Protocol:** Rules governing data communication (HTTP, TCP/IP, FTP)
- **Bandwidth:** Data transfer capacity of a network

Screenshot 2: URL Structure and Domain Name

3. URL and its Structure

`https://www.example.com:443/path/page.html?query=1`

Part	Example	Meaning
Protocol	https://	Method of data transfer
Domain	www.example.com	Website address
Port	:443	Communication endpoint
Path	/path/page.html	Resource location
Query	?query=1	Extra parameters

4. Domain Name

A domain name (e.g. *example.com*) is a human-readable address mapped to a numeric IP address via DNS (Domain Name System), so users don't need to remember IP numbers.

Screenshot 3: Web Browsers, Web Servers and Working Principle

5. Web Browsers and Web Servers

Web Browser: Client software (Chrome, Firefox) that requests and renders web pages.

Web Server: Software/hardware (Apache, Nginx) that stores and delivers web pages to browsers on request.

6. Working Principle of a Website

Browser (Client)

→ Request →
← Response ←

Web Server

1. User enters URL in browser → 2. Browser sends HTTP request to server → 3. Server processes request → 4. Server sends back HTML/CSS/JS → 5. Browser renders the page.

Screenshot 4: Client-side and Server-side Scripting

7. Client-side vs Server-side Scripting	
Client-side	Server-side
Runs in the browser	Runs on the web server
Languages: HTML, CSS, JavaScript	Languages: PHP, Python, Node.js
Faster, reduces server load	Handles database, authentication

CONCLUSION:

The webpage successfully explains the fundamental concepts of the Internet and World Wide Web. HTML5 is used to structure the webpage, while CSS3 is used to provide proper styling and visual presentation. Tables and diagrams make the concepts easier to understand. The webpage also demonstrates the basic communication between a web browser and a web server.

QUESTION 2:

Scenario: A college wants to digitize its student admission process.

Task: Design a responsive Student Admission Form using HTML5 and CSS3. Your webpage should demonstrate:

- **HTML Form Elements**
- **Different Input Types**
- **Media Elements (Image/Video/Audio)**
- **CSS3 Selectors**
- **Box Model**
- **Responsive User Interface**

ANSWER:

A responsive Student Admission Form is designed using HTML5 and CSS3 to collect important student details digitally. The form includes different input types such as text, email, telephone, date, radio buttons, dropdown list, range, file upload, textarea, and checkbox. CSS3 is used for styling, the box model, selectors, and responsive design.

SAMPLE CODE:

HTML CODE:

```
<!DOCTYPE html>

<h1>Student Admission Form</h1>

<form>

  <!-- Text input -->

  <div class="form-group">

    <label for="fname">Full Name</label>

    <input type="text" id="fname" name="fname"

      placeholder="Enter your full name" required>

  </div>

  <!-- Email input -->

  <div class="form-group">

    <label for="email">Email Address</label>

    <input type="email" id="email" name="email"

      placeholder="example@mail.com" required>

  </div>

  <!-- Telephone input -->

  <div class="form-group">

    <label for="phone">Phone Number</label>

    <input type="tel" id="phone" name="phone"

      pattern="[0-9]{10}"

      placeholder="10-digit number" required>

  </div>

  <!-- Date input -->

  <div class="form-group">

    <label for="dob">Date of Birth</label>

    <input type="date" id="dob" name="dob" required>

  </div>
```



```

<!-- Radio buttons -->
<div class="form-group">
    <label>Gender</label>

    <input type="radio" name="gender" value="male">
    <label class="inline">Male</label>

    <input type="radio" name="gender" value="female">
    <label class="inline">Female</label>

    <input type="radio" name="gender" value="other">
    <label class="inline">Other</label>
</div>

<!-- Dropdown list -->
<div class="form-group">
    <label for="course">Course</label>

    <select id="course" name="course" required>
        <option value="">-- Select Course --</option>
        <option>Computer Science Engineering</option>
        <option>Electronics & Communication</option>
        <option>Mechanical Engineering</option>
        <option>Civil Engineering</option>
    </select>
</div>

<!-- Range input -->
<div class="form-group">

```

CSS CODE:

```

/* Applies box sizing and common font to all elements */
display: inline;
font-weight: 400;
margin-right: 15px;

```

```
}  
  
/* Styles different form controls */  
  
input[type="text"],  
input[type="email"],  
input[type="tel"],  
input[type="date"],  
select,  
textarea {  
    width: 100%;  
    padding: 10px;  
    border: 1px solid #ccc;  
    border-radius: 6px;  
    font-size: 14px;  
}  
  
/* Styles the range input */  
  
input[type="range"] {  
    width: 100%;  
}  
  
/* Styles the submit button */  
  
button {  
    width: 100%;  
    padding: 12px;  
    background: #1a2b4c;  
    color: white;  
    border: none;  
    border-radius: 6px;  
    font-size: 16px;  
    cursor: pointer;  
}
```

```
/* Hover effect for the button */
button:hover {
    background: #2c5282;
}

/* Styles the campus media placeholder */
.campus {
    width: 100%;
    height: 180px;
    background: #2c5282;
    display: flex;
    align-items: center;
    justify-content: center;
    color: white;
    border-radius: 8px;
    font-size: 18px;
}

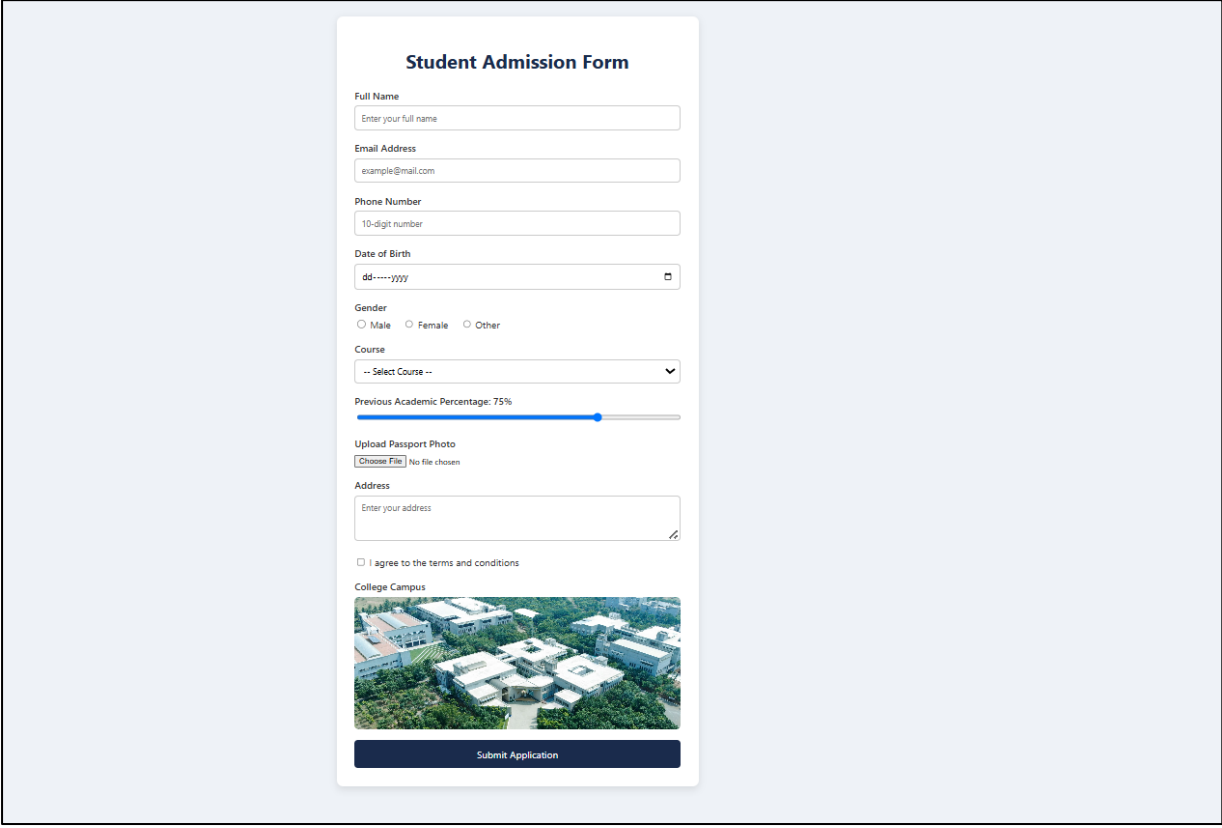
/* Responsive design for smaller screens */
@media (max-width: 480px) {
    .form-container {
        padding: 20px;
    }
    h1 {
        font-size: 22px;
    }
}
```

CODE EXPLANATION:

The HTML code creates the admission form using different HTML5 form elements such as text, email, telephone, date, radio buttons, dropdown, range, file upload, textarea, and checkbox. The CSS code uses selectors, the box model, spacing, colors, and media queries to make the form responsive on different screen sizes.

OUTPUT:

Screenshot 1 – Student Admission Form



The screenshot displays a 'Student Admission Form' on a light blue background. The form is a white card with a dark blue header containing the title 'Student Admission Form'. It includes several input fields: 'Full Name' (placeholder: 'Enter your full name'), 'Email Address' (placeholder: 'example@mail.com'), 'Phone Number' (placeholder: '10-digit number'), and 'Date of Birth' (placeholder: 'dd-....yyyy' with a calendar icon). Below these are radio buttons for 'Gender' (Male, Female, Other), a dropdown menu for 'Course' (placeholder: '-- Select Course --'), and a range slider for 'Previous Academic Percentage' (set to 75%). There is a file upload section for 'Upload Passport Photo' with a 'Choose File' button and 'No file chosen' text. An 'Address' field (placeholder: 'Enter your address') with a location pin icon follows. A checkbox for 'I agree to the terms and conditions' is present. At the bottom, there is a 'College Campus' image showing a modern university building and a dark blue 'Submit Application' button.

Screenshot 2 – Form with Input Elements and Media Section

Student Admission Form

Full Name
SUGHANTH

Email Address
sugu123@gmail.com

Phone Number
9876543210

Date of Birth
14-Jun-2007

Gender
☒ Male ☐ Female ☐ Other

Course
Computer Science Engineering

Previous Academic Percentage: 91%

Upload Passport Photo
Choose File | passport photo.png

Address
111, Perundurai, Erode

☒ I agree to the terms and conditions

College Campus

Submit Application

CONCLUSION:

The Student Admission Form successfully demonstrates HTML5 form elements and different input types. CSS3 selectors and the box model are used to create a clean and organized interface. A responsive media query ensures that the form adapts to smaller screen sizes.

QUESTION 3

Scenario: An online fashion store wants an attractive home page to improve customer engagement.

Task: Design the homepage using HTML5 and CSS3 by incorporating:

- Backgrounds and Borders
- Text Effects
- CSS Animations
- Multiple Column Layout
- User Interface Design Principles

ANSWER:

An attractive homepage helps an online fashion store present its products and services effectively to customers. HTML5 is used to structure the webpage into sections such as navigation, hero banner, product cards, and footer. CSS3 is used to apply backgrounds, borders, text effects, animations, and multiple-column layouts.

The webpage designed for this question represents a fashion store named **Vogue Threads** with a stylish hero section, navigation menu, promotional strip, product cards, and responsive layout.

SOURCE CODE:

HTML CODE:

```
<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8">

  <title>Vogue Threads</title>

  <link rel="stylesheet" href="q3_style.css">

</head>

<body>

  <!-- Navigation and hero section -->

  <header class="hero">

    <nav>

      <b>VOGUE THREADS</b>

      <span>Home | Collections | Sale | Contact</span>

    </nav>

    <h1>Redefine Your Style</h1>

    <p>Discover the latest fashion trends</p>

    <button>Shop Now</button>

  </header>

  <!-- Promotional animation -->

  <div class="offer">
```

◆ FREE SHIPPING ◆ 30% OFF ON FIRST ORDER ◆

</div>

<!-- Multiple column layout -->

<section class="content">

<h2>Trending Now</h2>

<div class="products">

<!-- Product cards -->

<div class="card"> 
Summer Dress
₹1,499</div>

<div class="card"> 
Denim Jacket
₹2,299</div>

<div class="card"> 
Casual Sneakers
₹1,899</div>

</div>

</section>

<footer>

© 2026 Vogue Threads

</footer>

</body>

</html>

CSS CODE:

/* Basic page styling */

* {

box-sizing: border-box;

font-family: Arial;

}

body {

margin: 0;

text-align: center;

}

/* Background and border */

```
.hero {  
    background: linear-gradient(135deg, #6a0572, #ff6b6b);  
    color: white;  
    padding: 40px;  
    border-bottom: 5px solid #ff6b6b;  
}  
  
/* Navigation layout */  
nav {  
    display: flex;  
    justify-content: space-between;  
}  
  
/* Text effect and animation */  
h1 {  
    font-size: 45px;  
    text-shadow: 0 0 20px white;  
    animation: glow 2s infinite alternate;  
}  
  
/* CSS animation */  
@keyframes glow {  
    from { text-shadow: 0 0 10px white; }  
    to { text-shadow: 0 0 30px #ffd6e0; }  
}  
  
/* Button styling */  
button {  
    padding: 12px 30px;  
    border: none;  
    border-radius: 20px;  
    background: white;  
    color: #6a0572;
```



```
}  
  
/* Promotional animation */  
  
.offer {  
    background: #1a2b4c;  
    color: white;  
    padding: 12px;  
    animation: move 5s infinite;  
}  
  
@keyframes move {  
    from { transform: translateX(10%); }  
    to { transform: translateX(-10%); }  
}  
  
/* Multiple column layout */  
  
.products {  
    display: flex;  
    justify-content: center;  
    gap: 20px;  
    padding: 30px;  
}  
  
/* Product card UI */  
  
.card {  
    width: 200px;  
    padding: 30px;  
    background: #f8f9fa;  
    border: 1px solid #ddd;  
    border-radius: 10px;  
    box-shadow: 0 4px 10px #ccc;  
}
```

```

/* Hover effect */
.card:hover {
    transform: translateY(-8px);
}

footer {
    background: #1a2b4c;
    color: white;
    padding: 15px;
}

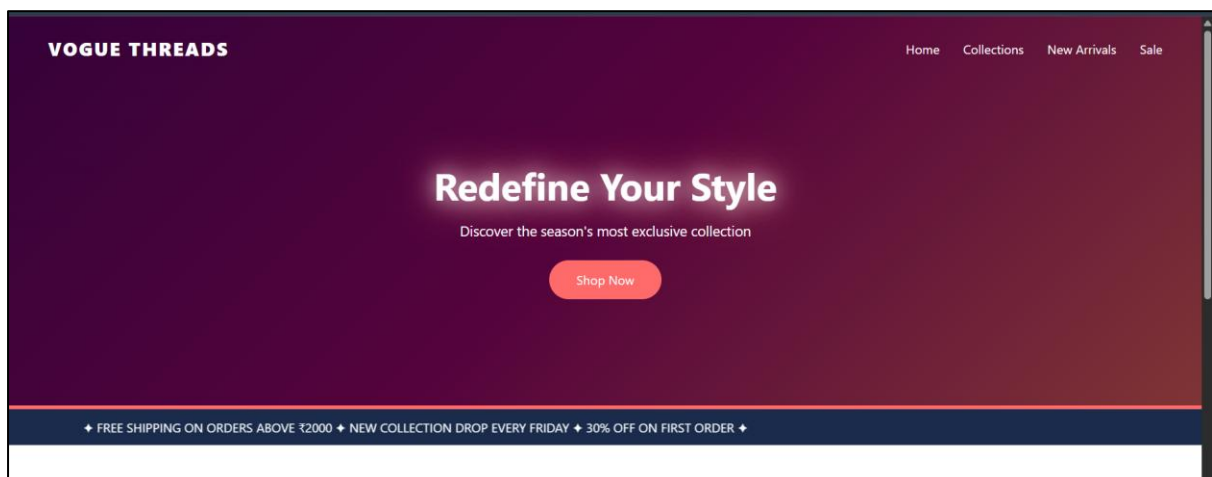
```

CODE EXPLANATION:

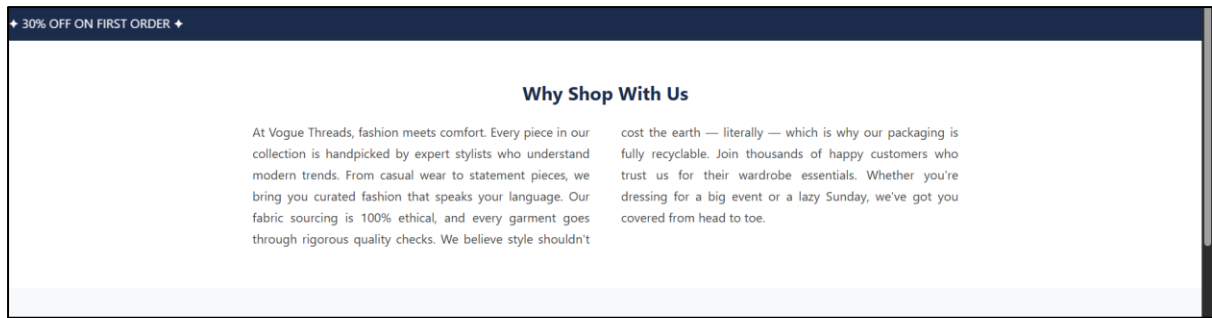
The HTML code creates a fashion store homepage with navigation, hero section, promotional content, and product cards. The CSS code demonstrates backgrounds and borders, text effects, CSS animations, multiple-column layout, hover effects, and basic user interface design principles.

OUTPUT SCREENSHOTS:

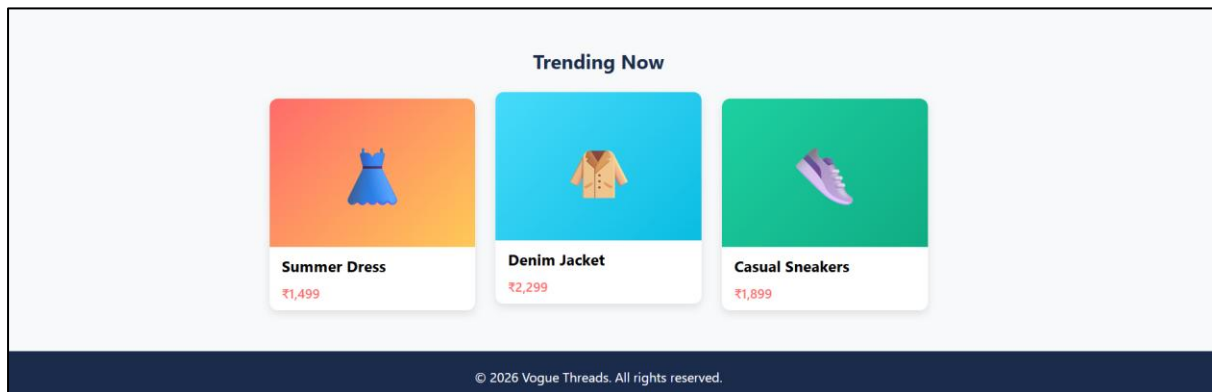
Screenshot 1: Fashion Store Homepage and Navigation



Screenshot 2: Hero Section with Text Effects and Animation



Screenshot 3: Multiple Column Layout and Trending Products



CONCLUSION:

The fashion store homepage was successfully designed using HTML5 and CSS3. Backgrounds, borders, text effects, animations, multiple-column layouts, product cards, and responsive design were implemented to create an attractive and user-friendly interface. The design improves customer engagement by providing clear navigation and visually appealing product presentation.

QUESTION 4:

Scenario: A school wants to automate the student grade calculation process.

Task: Develop a JavaScript program that accepts marks of five subjects and displays the Total, Average, Grade, and Pass/Fail status. The solution should demonstrate:

- Variables and Data Types
- Literals
- Operators
- Statements (Selection and Iteration)
- User-defined Functions

ANSWER:

A student grade calculator can be used to automatically process marks and determine the academic performance of a student. JavaScript is used to accept the marks of five subjects, calculate the total and average, assign a grade, and determine the pass/fail status.

The webpage created for this question demonstrates JavaScript variables and data types, literals, arithmetic and comparison operators, selection statements, iteration using a for loop, and a user-defined function.

SOURCE CODE:

HTML CODE:

```
<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8">

  <title>Student Grade Calculator</title>

  <style>

    /* Basic page styling */

    body {

      font-family: Arial;

      max-width: 500px;

      margin: 40px auto;

      background: #f0f4f8;

      padding: 20px;

    }

    .container {

      background: white;

      padding: 25px;

      border-radius: 10px;

    }

    input, button {

      width: 100%;
```

```
padding: 10px;
margin: 8px 0;
box-sizing: border-box;
}
button {
background: #1a2b4c;
color: white;
border: none;
cursor: pointer;
}
#result {
margin-top: 15px;
padding: 15px;
background: #eef2f7;
}
</style>
</head>
<body>
<div class="container">
  <!-- Page heading -->
  <h2>Student Grade Calculator</h2>
  <!-- Accepts marks for five subjects -->
  <input type="number" id="sub1" placeholder="Subject 1 Marks">
  <input type="number" id="sub2" placeholder="Subject 2 Marks">
  <input type="number" id="sub3" placeholder="Subject 3 Marks">
  <input type="number" id="sub4" placeholder="Subject 4 Marks">
  <input type="number" id="sub5" placeholder="Subject 5 Marks">
  <!-- Calls the user-defined JavaScript function -->
  <button onclick="calculateGrade()">Calculate Grade</button>
```

```
<!-- Displays the calculated result -->
<div id="result"></div>
</div>
<script src="q4_script.js"></script>
</body>
</html>
```

JAVASCRIPT CODE:

```
// User-defined function for grade calculation
function calculateGrade() {
    // Variables storing subject marks
    let marks = [
        Number(document.getElementById("sub1").value),
        Number(document.getElementById("sub2").value),
        Number(document.getElementById("sub3").value),
        Number(document.getElementById("sub4").value),
        Number(document.getElementById("sub5").value)
    ];
    // Selection statement to validate marks
    for (let i = 0; i < marks.length; i++) {
        if (marks[i] < 0 || marks[i] > 100) {
            alert("Enter marks between 0 and 100.");
            return;
        }
    }
    // Arithmetic operators calculate total and average
    let total = 0;
    for (let i = 0; i < marks.length; i++) {
        total = total + marks[i];
    }
}
```

```
}  
  
let average = total / 5;  
  
// Selection statement determines grade  
  
let grade;  
  
if (average >= 90)  
    grade = "A+";  
  
else if (average >= 80)  
    grade = "A";  
  
else if (average >= 70)  
    grade = "B";  
  
else if (average >= 60)  
    grade = "C";  
  
else if (average >= 40)  
    grade = "D";  
  
else  
    grade = "F";  
  
// Determines pass or fail  
  
let status = "Pass";  
  
for (let i = 0; i < marks.length; i++) {  
    if (marks[i] < 40) {  
        status = "Fail";  
        break;  
    }  
}  
  
// Displays the result  
  
document.getElementById("result").innerHTML =  
    "<b>Total:</b> " + total + " / 500<br>" +  
    "<b>Average:</b> " + average.toFixed(2) + "%<br>" +  
    "<b>Grade:</b> " + grade + "<br>" +
```

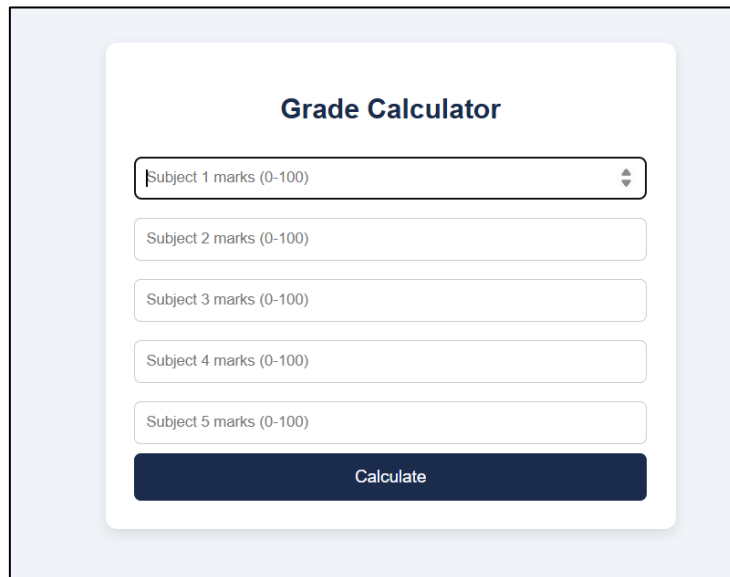
```
"<b>Status:</b> " + status;  
}
```

CODE EXPLANATION:

The HTML code provides input fields for five subject marks and a button to calculate the result. The JavaScript code uses variables and numeric data types, literals, arithmetic and comparison operators, if-else selection statements, for loop iteration, and a user-defined function to calculate and display the total, average, grade, and pass/fail status.

OUTPUT SCREENSHOTS:

Screenshot 1: Student Grade Calculator Input Form

A screenshot of a web form titled "Grade Calculator". The form is centered on a light blue background. It contains five input fields, each labeled "Subject 1 marks (0-100)" through "Subject 5 marks (0-100)". The first field has a small upward and downward arrow icon on its right side. Below the input fields is a dark blue button with the text "Calculate" in white.

Screenshot 2: Grade Calculation – Pass Result

Grade Calculator

Calculate

Total Marks: 460 / 500

Average: 92.00%

Grade: A+

Status: **Pass**

Screenshot 3: Grade Calculation – Fail Result

Grade Calculator

Calculate

Total Marks: 191 / 500

Average: 38.20%

Grade: F

Status: **Fail**

CONCLUSION:

The Student Grade Calculator was successfully developed using HTML and JavaScript. The program accepts marks for five subjects and automatically calculates the total, average, grade, and pass/fail status. It demonstrates important JavaScript concepts including variables, data types, literals, operators, selection statements, iteration, and user-defined functions.

QUESTION 5

Scenario: A supermarket wants to generate customer bills automatically.

Task: Develop a JavaScript program to calculate the bill amount for three purchased products. The program should:

- Accept quantity and unit price for each product
- Calculate the total bill
- Apply a 10% discount if the total amount exceeds ₹2000
- Display the final payable amount

The solution should demonstrate:

- Variables and Data Types
- Literals
- Operators
- Statements
- User-defined Functions

ANSWER:

The **Supermarket Billing System** is a JavaScript-based application that calculates the bill for three purchased products. It accepts the quantity and unit price of each product, calculates the individual product costs and total bill, and automatically applies a 10% discount when the total exceeds ₹2000.

The program demonstrates variables and data types, literals, arithmetic and comparison operators, selection statements using if, and a user-defined function.

SOURCE CODE:

HTML CODE:

```
<!DOCTYPE html>

<html lang="en">

<head>

  <!-- Defines character encoding -->

  <meta charset="UTF-8">

  <title>Supermarket Billing System</title>

  <style>

    /* Basic page styling */
```

```
body {  
    font-family: Arial;  
    background: #eef2f7;  
    max-width: 500px;  
    margin: 40px auto;  
    padding: 20px;  
}  
  
.container {  
    background: white;  
    padding: 25px;  
    border-radius: 10px;  
}  
  
h2 {  
    text-align: center;  
    color: #1a2b4c;  
}  
  
.product {  
    margin-bottom: 15px;  
}  
  
input {  
    width: 47%;  
    padding: 9px;  
    margin: 5px 1%;  
    box-sizing: border-box;  
}  
  
button {  
    width: 100%;  
    padding: 12px;  
    background: #1a2b4c;
```

```
        color: white;

        border: none;

        cursor: pointer;
    }

    #result {

        margin-top: 20px;

        padding: 15px;

        background: #eef2f7;

    }
</style>
</head>
<body>
<div class="container">

    <!-- Page heading -->

    <h2>🛒 Supermarket Billing System</h2>

    <!-- Product 1 -->

    <div class="product">

        <b>Product 1</b><br>

        <input type="number" id="qty1" placeholder="Quantity">

        <input type="number" id="price1" placeholder="Unit Price">

    </div>

    <!-- Product 2 -->

    <div class="product">

        <b>Product 2</b><br>

        <input type="number" id="qty2" placeholder="Quantity">

        <input type="number" id="price2" placeholder="Unit Price">

    </div>

    <!-- Product 3 -->

    <div class="product">
```

```

        <b>Product 3</b><br>
        <input type="number" id="qty3" placeholder="Quantity">
        <input type="number" id="price3" placeholder="Unit Price">
    </div>

    <!-- Calls the user-defined function -->
    <button onclick="calculateBill()">Generate Bill</button>

    <!-- Displays the final bill -->
    <div id="result"></div>
</div>

<!-- Links the JavaScript file -->
<script src="q5_script.js"></script>
</body>
</html>

```

JAVASCRIPT CODE:

```

// User-defined function to calculate the bill
function calculateBill() {
    // Variables storing quantity and unit price
    let qty1 = Number(document.getElementById("qty1").value);
    let price1 = Number(document.getElementById("price1").value);
    let qty2 = Number(document.getElementById("qty2").value);
    let price2 = Number(document.getElementById("price2").value);
    let qty3 = Number(document.getElementById("qty3").value);
    let price3 = Number(document.getElementById("price3").value);

    // Selection statement for validation
    if (qty1 < 0 || price1 < 0 ||
        qty2 < 0 || price2 < 0 ||
        qty3 < 0 || price3 < 0) {
        alert("Please enter valid values.");
    }
}

```

```

        return;
    }

    // Calculate individual product costs
    let cost1 = qty1 * price1;
    let cost2 = qty2 * price2;
    let cost3 = qty3 * price3;

    // Calculate total bill using addition operator
    let totalBill = cost1 + cost2 + cost3;

    // Apply 10% discount if bill exceeds ₹2000
    let discount = 0;
    if (totalBill > 2000) {
        discount = totalBill * 0.10;
    }

    // Calculate final payable amount
    let finalAmount = totalBill - discount;

    // Display the bill
    document.getElementById("result").innerHTML = `
        <b>Product 1:</b> ₹${cost1.toFixed(2)}<br>
        <b>Product 2:</b> ₹${cost2.toFixed(2)}<br>
        <b>Product 3:</b> ₹${cost3.toFixed(2)}<br><br>
        <b>Total Bill:</b> ₹${totalBill.toFixed(2)}<br>
        <b>Discount:</b> ₹${discount.toFixed(2)}<br>
        <b>Final Payable Amount:</b> ₹${finalAmount.toFixed(2)} `;
    }

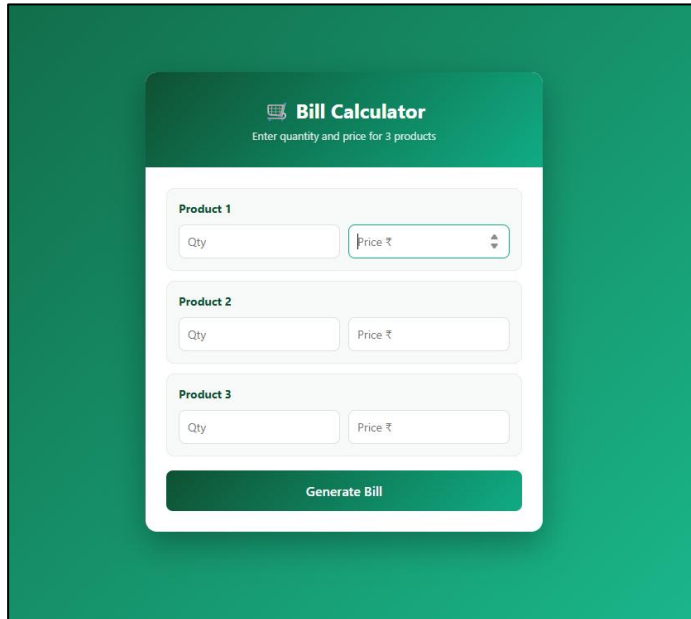
```

CODE EXPLANATION:

The HTML code provides input fields for the quantity and unit price of three products. The JavaScript code uses variables, numeric data types, literals, arithmetic and comparison operators, an if selection statement, and a user-defined function to calculate the product costs, total bill, 10% discount, and final payable amount.

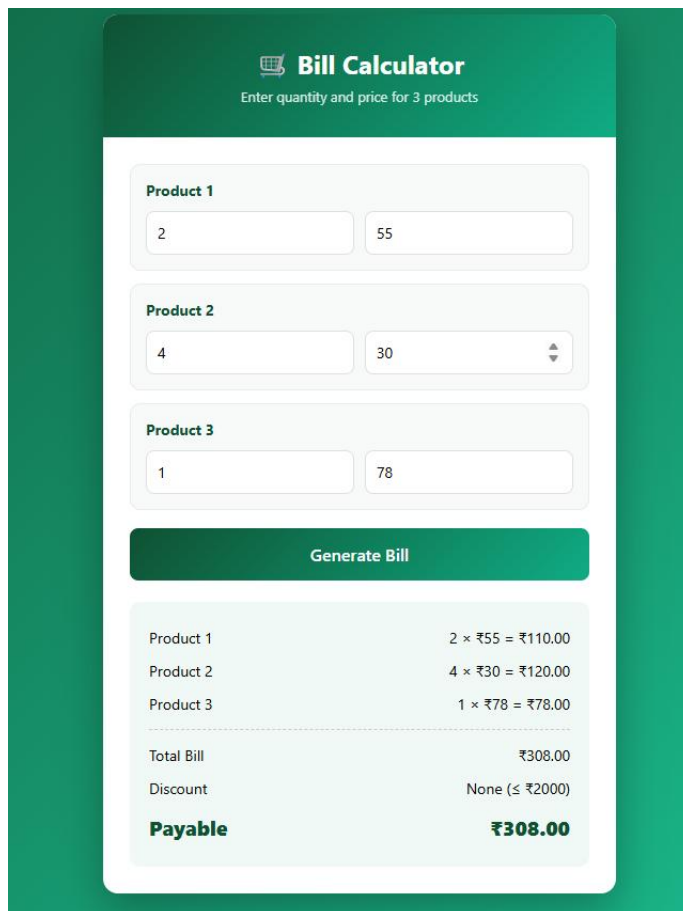
OUTPUT SCREENSHOTS:

Screenshot 1: Supermarket Billing Input Form



The screenshot shows a web application titled "Bill Calculator" with the subtitle "Enter quantity and price for 3 products". The interface is set against a dark green background. It features three input sections for Product 1, Product 2, and Product 3. Each section contains a "Qty" input field and a "Price ₹" input field with a dropdown arrow. At the bottom of the input section is a green button labeled "Generate Bill".

Screenshot 2: Final Payable Amount without 10% Discount



The screenshot shows the same "Bill Calculator" application, but now with data entered in the input fields. The "Generate Bill" button has been clicked, and the results are displayed in a light green box at the bottom. The results show the calculation for each product and the total bill.

Product 1	2 × ₹55 = ₹110.00
Product 2	4 × ₹30 = ₹120.00
Product 3	1 × ₹78 = ₹78.00
Total Bill	₹308.00
Discount	None (≤ ₹2000)
Payable	₹308.00

Screenshot 3: Final Payable Amount without 10% Discount

Bill Calculator

Enter quantity and price for 3 products

Product 1

10

200

Product 2

4

400

Product 3

1

500

Generate Bill

Product 1

10 × ₹200 = ₹2000.00

Product 2

4 × ₹400 = ₹1600.00

Product 3

1 × ₹500 = ₹500.00

Total Bill

₹4100.00

Discount (10%)

-₹410.00

Payable

₹3690.00

CONCLUSION:

The Supermarket Billing System was successfully developed using HTML and JavaScript. The program accepts the quantity and unit price of three products, calculates the total bill, applies a 10% discount when the bill exceeds ₹2000, and displays the final payable amount. The program demonstrates variables, data types, literals, operators, statements, and user-defined functions.